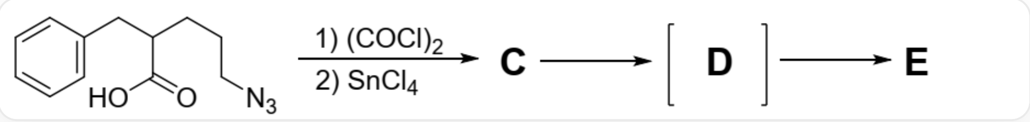
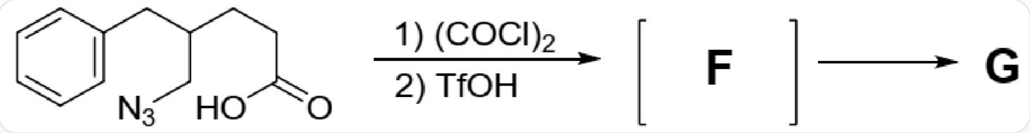
Question

In situ activation of ω-azidocarboxylic acids can induce some interesting reactions. 5-Azido-2-benzylpentanoic acid reacts with oxalyl chloride and tin(IV) chloride to form intermediate $\mathbf{C}$, which subsequently undergoes two-step reactions to yield intermediate $\mathbf{D}$ and product $\mathbf{E}$.



The image shows a multistep reaction: O=C(O)C(CCCN=[N+]=[N-])CC1=CC=CC=C1>ClC(C(Cl)=O)=O.Cl[Sn](Cl)(Cl)Cl>[**C**],[**C**]>[**D**],[**D**]>[**E**] 5-Azido-2-benzylpentanoic acid reacts with oxalyl chloride and tin(IV) chloride to form intermediate C, which subsequently undergoes two-step reactions to yield intermediate D and product E

A slight structural modification of the substrate yields 5-azido-4-benzylpentanoic acid. This compound reacts with oxalyl chloride and trifluoromethanesulfonic acid to form the final intermediate $\mathbf{F}$, which then undergoes one more reaction to produce product $\mathbf{G}$, distinct from $\mathbf{E}$.



The image shows a multistep reaction: O=C(O)CCC(CN=[N+]=[N-])CC1=CC=CC=C1>ClC(C(Cl)=O)=O.O=S(C(F)(F)F)(O)=O>[**F**],[**F**]>[**G**] 5-Azido-4-benzylpentanoic acid reacts with oxalyl chloride and trifluoromethanesulfonic acid to form the final intermediate F, which then undergoes one more reaction to produce product G

None of the intermediates contain Cl. The following statements are correct:

A. The intermediate $\mathbf{C}$ contains a carboxyl group

B. The intermediate **D** is a rearrangement product containing two six-membered rings.

C. The product **E** is a secondary amine.

D. The intermediate **F** has an isocyanate structure.

E. The product **G** contains three rings, with the carbonyl group located on the five-membered ring.

F. The second reaction can form two different rearrangement intermediates, among which the isocyanate cation intermediate reacts with the benzene ring to yield a more stable product.

G. None of the above options are correct

Answer

Correct Answer: E

Detailed Explanation

In the first reaction, the azide and carboxylic acid undergo intramolecular cyclization to yield intermediate **C**,

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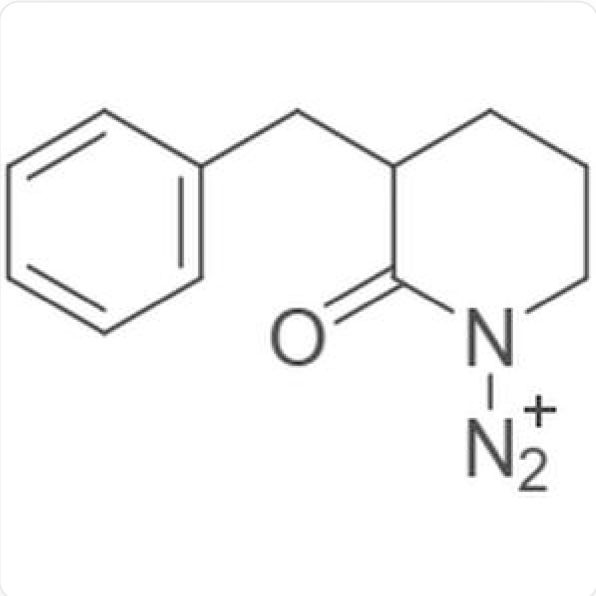
The azido group and carboxylic acid undergo intramolecular cyclization to yield intermediate **C**, A is incorrect

The structural formula of intermediate **C** is O=C1N([N+]#N)CCCC1CC2=CC=CC=C2,



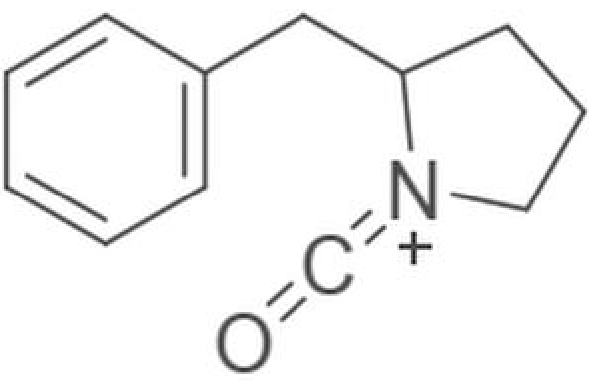
O=C1N([N+]#N)CCCC1CC2=CC=CC=C2

Intermediate $\mathbf{C}$ undergoes rearrangement to form intermediate $\mathbf{D}$, which has an isocyanate structure containing a five-membered ring,

CHECKPOINT 1 PTS

Intermediate $\mathbf{D}$ has an isocyanate structure containing a five-membered ring, B is incorrect

The structural formula of intermediate $\mathbf{D}$ is O=C=[N+]1CCCC1CC2=CC=CC=C2,



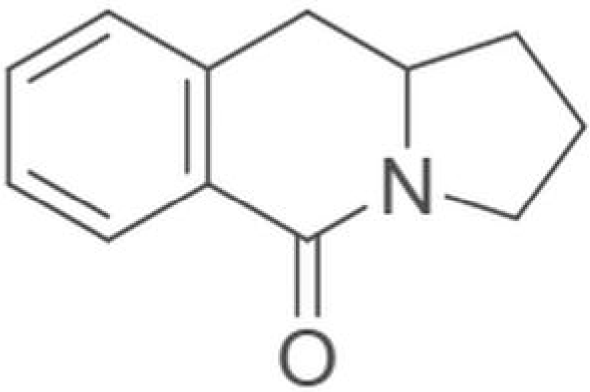
O=C=[N+]1CCCC1CC2=CC=CC=C2

The benzene ring reacts with the isocyanate, undergoing intramolecular cyclization to yield product $\mathbf{E}$. Since water is not mentioned in the reaction conditions, hydrolysis to produce carbon dioxide and a secondary amine does not occur,

CHECKPOINT 1 PTS

The benzene ring reacts with the isocyanate to yield product $\mathbf{E}$. Since water is not mentioned in the reaction conditions, a secondary amine is not obtained, C is incorrect

The structure of product $\mathbf{E}$ is C1=CC2=C(C=C1)CN3C(CCC3=O)C2C1=CC2=C(C=C1)C(=O)N3CCCC3C2,



C1=CC2=C(C=C1)C(=O)N3CCCC3C2

In the second reaction, the azide and carboxylic acid react and undergo rearrangement to form the imine intermediate $\mathbf{F}$,

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Intermediate $\mathbf{F}$ has an imine structure, D is incorrect

The difference between product $\mathbf{E}$ and product $\mathbf{G}$ lies in the preferential migration of the more substituted carbon atom during the rearrangement,

CHECKPOINT 1 PTS

The difference between product **E** and product **G** lies in the preferential migration of the more substituted carbon atom during the rearrangement

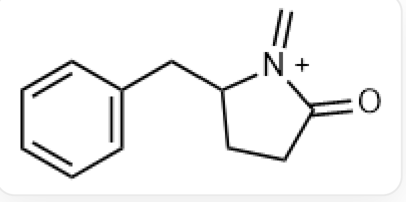
Since nitrogen gas is released in this reaction, the reaction is irreversible, and G is incorrect,

CHECKPOINT 1 PTS

Since nitrogen gas is released in this reaction, the reaction is irreversible

Option F is incorrect.

The structure of intermediate **F** is C=C1CCC([N+]1=C)CC2=CC=CC=C2,



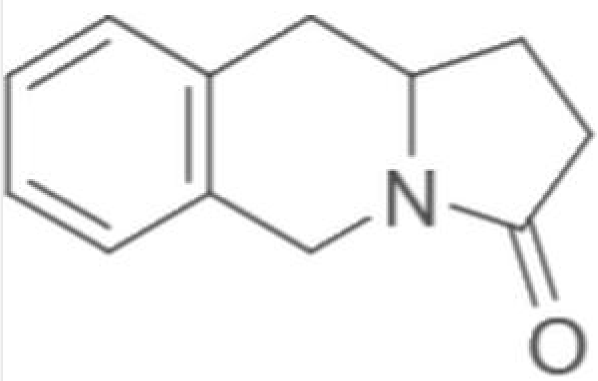
O=C1CCC([N+]1=C)CC2=CC=CC=C2

The benzene ring reacts with the imine, undergoing intramolecular cyclization to ultimately yield product **G**. Product **G** has three rings, and the carbonyl group is located on the five-membered ring,

CHECKPOINT 1 PTS

Product **G** has three rings, and the carbonyl group is located on the five-membered ring, so E is correct

The structure of product **G** is O=C1CCC([N+]1=C)CC2=CC=CC=C2,



C1=CC2=C(C=C1)CN3C(CCC3=O)C2